

KEY PARTNERS - EBS (Energiebedrijven Suriname) — tariff data, potential rollout partner - Licensed electricians — panel installation network - Local hardware importers (ESP32, sensors, clamps) - Telesur / Digicel — connectivity for IoT data - ADeKUS — engineering talent, credibility, pilot households - Google (Gemini API) — AI advisory layer, model-agnostic by design	KEY ACTIVITIES - Hardware assembly & per-unit calibration - Firmware & platform development (offline-first) - AI advisory: tips in Dutch & Sranan Tongo - Tiered EBS billing engine & bill prediction - Installation coordination with electricians	VALUE PROPOSITIONS - See your electricity bill BEFORE it arrives — tier-aware prediction - Advice in your own language: Dutch & Sranan Tongo - Built for Suriname's unstable grid: outage & brownout logging, offline-first - Know which tariff tier you are in and how to stay under the next one - Budget mode: set a monthly limit, get warned before you exceed it - Evidence of grid quality (outage history) households can act on - No smart meter required — clips onto existing breaker panel - Future: per-appliance insight via AI fingerprinting	CUSTOMER RELATIONSHIPS - Self-service dashboard (web app) - AI advisor as ongoing personal touchpoint - Installation moment = trust-building visit - Community word-of-mouth (bill savings are visible)	CUSTOMER SEGMENTS - PRIMARY: Surinamese households on EBS household tariff, bill-sensitive, hit by tier jumps - Households with frequent outages wanting grid evidence - Budget-conscious families (subsidy-band households) - SECONDARY: small businesses (niet-huishoudelijk flat 4.820/kWh — simpler billing, price-sensitive) - FUTURE: groot verbruikers — time-of-use, kVA demand & power-factor optimisation (measurable ROI, no consumption cut needed)
	KEY RESOURCES - Proprietary firmware: outage-resilient logging, timestamp reconstruction - Tariff-aware billing engine (EBS block pricing) - Sensor pipeline: current clamps + voltage sensing - Growing dataset of Surinamese household usage - Bilingual AI prompt & advisory system		CHANNELS - Direct sales via landing page & social media - Electrician referral network - Hardware shops as resellers - Future: EBS partnership channel	
COST STRUCTURE - Per-unit hardware BOM: ESP32, ADS1115, SCT-013 clamp(s), ZMPT101B, PSU, enclosure - Installation labour (electrician fee per unit) · Cloud & database hosting (Supabase) - AI API usage (batched, ~fractions of a cent per advisory) · Support & calibration			REVENUE STREAMS - Device sale (one-time, hardware + installation) - Monthly subscription: AI advisory, bill prediction, outage history - FUTURE: commercial tier for businesses (demand & power-factor analytics, premium pricing) - FUTURE: anonymised consumption insights for utility planning (with consent)	

Notes, Assumptions & Open Questions

Why this problem, why here

- EBS household billing is tiered (1.971 / 2.850 / 3.330 / 4.680 SRD per kWh, Dec 2024 sheet). Crossing a block raises the marginal price of every further kWh by up to 45%. Households cannot see this coming; beepBoop makes the tier position visible and predicts the bill.
- Grid instability is normal life in Suriname: the prototype has already logged real brownouts (down to 103 V) and outages during bench testing. Outage history is evidence households currently have no way to capture.
- Advisory in Sranan Tongo is a differentiator no imported product (Sense, Emporia) offers.

What is proven vs. assumed

- PROVEN in prototype: calibrated current & voltage sensing, tiered billing engine, offline-first logging with timestamp reconstruction, outage/brownout detection, bilingual AI advisory grounded in measured data.
- ASSUMED, to validate: willingness to pay (device price & subscription level unset), installation cost per unit, subsidy band mechanics (placeholder 100 SRD in current build), whether Dec 2024 tariffs still hold.

Unit economics (to be filled with real quotes)

- BOM target: components are commodity parts; production would replace the ADS1115 with a metering IC (e.g. ATM90E32) for multi-circuit and power-factor accuracy. Pricing decision pending importer quotes.
- AI cost per user is negligible at batched-advisory volumes (fractions of SRD 0.01 per tip on free/flash-tier models); the system is model-agnostic (Gemini today, swappable).

Scaling path

- Households first (this canvas). Small business second: flat 4.820/kWh makes billing simpler than households.
- Groot verbruikers third: time-of-use tariffs, kVA demand charges and cosphi penalties mean businesses can cut bills without cutting consumption — load shifting, peak flattening, power-factor correction. Requires simultaneous-sampling metering hardware; a deliberate v2.